

COMP5331 Project Proposal Comment

Group No.: 15

Title: Stress-Testing Simple Graph Features under Controlled Link-Anomaly Injection

Type: Research

Comment (given by Raymond):

1. This is a research project (Group 15) that stress-tests the effectiveness of simple historical-graph features (the TGF detector) under controlled link-anomaly injection. The group observes that randomly injected edges may introduce artifacts that make anomaly detection artificially easy. Keeping the dataset, feature extractor, classifier family and evaluation protocol fixed, they will compare standard random link injection against an activity-matched alternative that preserves the recent activity profile of the original endpoints. The study will quantify the resulting change in detection performance and identify which feature groups are responsible for any performance drop. The intended contribution is a reproducible boundary test of the conditions under which performance based on trivial graph features remains valid, rather than the introduction of a new detector or anomaly taxonomy.
2. Reproduce the original TGF pipeline on the chosen dataset(s) under pure random injection and verify that the published performance figures can be matched. Immediately implement the activity-matched injection generator behind a common interface that enforces a strict pre-arrival history protocol (only interactions with $\text{timestamp} < t$ are visible). Confirm that the activity-matched generator actually reduces the intended discrepancy (recent activity profiles of the endpoints) before running the full comparative evaluation and feature-group ablation.
3. Matching only activity profiles does not guarantee that the injected edges are “harder” in every other structural sense. Residual differences in degree, local neighborhood or temporal burstiness may still leak into the features. The group should therefore report diagnostic statistics that quantify how well each generator matches the observed distribution of normal edges. Finally, any claim about “evaluation bias” should be supported by concrete performance deltas and feature-importance evidence rather than by the mere existence of a performance gap.
4. The official TGF implementation may be used unchanged solely for the reproduction baseline. Any reused fragments must be explicitly acknowledged. All new components—the activity-matched anomaly generator, the diagnostic checks that verify profile matching, the feature-group ablation harness, and the comparative evaluation pipeline—must be implemented independently by the group. Substantial copying of public or author-released code beyond the declared reproduction baseline without proper attribution constitutes plagiarism.